Appendix: Low-Rank Path Constraints, CPWL Face Moments, and Fourier Scaling Laws

Anonymous Workshop Submission

1 Why the path-count wording is wrong but useful

The informal slogan is "low rank means fewer paths." Strictly speaking this is false. A dense matrix $W \in \mathbb{R}^{m \times n}$ has one edge between each input and output channel. A factorization $W = UV^\top$ inserts a latent rank coordinate, so the computational graph can contain $nr + mr$ edges and many two-hop graph paths. What decreases is the dimension of the set of possible coefficients attached to paths.

For one factorized layer,

$$W_{ij} = \sum_{a=1}^{r} U_{ia} V_{ja}. \tag{1}$$

A path coefficient involving this edge is no longer independently indexed by (i, j); it factors through $a \in \{1, \dots, r\}$. With L factorized layers, the full active path tensor belongs to a tensor network whose bond dimensions are the ranks. This is the precise version used in the main text.

2 Proof of the normal-subspace constraint

Fix a region R_ε of the first $\ell - 1$ layers. On that region,

$$h_{\ell-1}(x) = B_{\ell-1,\varepsilon} x + c_{\ell-1,\varepsilon}. \tag{2}$$

For layer ℓ with $W_\ell = U_\ell V_\ell^\top$, the preactivation of neuron i is

$$\begin{aligned} a_{\ell i}(x) &= e_i^\top U_\ell V_\ell^\top (B_{\ell-1,\varepsilon} x + c_{\ell-1,\varepsilon}) + b_{\ell i} \qquad (3)\\ &= u_{\ell i}^\top V_\ell^\top B_{\ell-1,\varepsilon} x + \text{const}. \qquad (4) \end{aligned}$$

Thus the normal is $n_{\ell i \varepsilon} = B_{\ell-1,\varepsilon}^\top V_\ell u_{\ell i}$ and belongs to $\operatorname{Im}(B_{\ell-1,\varepsilon}^\top V_\ell)$. Moreover

$$\operatorname{rank}(B_{\ell,\varepsilon}) = \operatorname{rank}(D_{\ell,\varepsilon} U_\ell V_\ell^\top B_{\ell-1,\varepsilon}) \le \min\{\operatorname{rank}(B_{\ell-1,\varepsilon}), r_\ell\}. \tag{5}$$

Induction from $\operatorname{rank}(B_0) = d$ gives

$$\dim \operatorname{Im}(B_{\ell-1,\varepsilon}^\top V_\ell) \le \min(d, r_1, \dots, r_\ell). \tag{6}$$

This is the geometric bottleneck behind the face-moment proxy.

3 From arrangement counts to face moments

Let m hyperplanes be in general position inside a ρ-dimensional normal subspace. A codimension-q face can only be created by selecting q active hyperplanes and then counting regions of the induced arrangement on the intersection. The conservative proxy

$$N_q(m, \rho) \le \binom{m}{q} \sum_{j=0}^{\rho-q} \binom{m-q}{j} \tag{7}$$

therefore suffices for our purpose. It has the key qualitative property $N_q(m, \rho) = 0$ when $q > \rho$. In particular, rank constraints suppress high codimension first.

The Fourier-relevant moment is not just a face count. It weighs each face by jump sizes and volumes:

$$M_q(f) = \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \operatorname{vol}_{d-q}(F)^2 \omega_F. \tag{8}$$

The count proxy controls the combinatorial part of M_q, while rank also affects jump magnitudes. The experiments in the main paper use multi-way activation junctions as a measurable proxy for the high-q part of M_q.

4 Fourier shell expansion details

For a compactly windowed CPWL function $g = \chi f$, repeated integration by parts gives boundary terms on the faces where derivatives jump. A schematic form is

$$\widehat{g}(\xi) = \sum_{q=1}^{d} \sum_{F \in \mathcal{F}_q} \frac{e^{-i\langle \xi, x_F \rangle}}{\|\xi\|^{q+1}} A_F(\xi/\|\xi\|) + O(\|\xi\|^{-d-2}). \tag{9}$$

Squaring and averaging over shells removes many oscillatory cross terms, leading to

$$S_f(\rho) \lesssim \sum_{q=1}^{d} M_q(f) \rho^{-2(q+1)}. \tag{10}$$

The effective slope is a weighted average of the exponents $2(q + 1)$:

$$\alpha_{\text{eff}}(\rho) = \frac{\sum_q 2(q+1) M_q \rho^{-2(q+1)}}{\sum_q M_q \rho^{-2(q+1)}}. \tag{11}$$

Therefore, removing $q \geq 2$ terms makes the observed slope smaller in magnitude. This is the mathematical version of the "better Fourier scaling" claim.

5 The special 1D case

A 1D ReLU network is a spline. If f' has jumps Δ_j at knots t_j, then in distributions

$$f'' = \sum_j \Delta_j \delta_{t_j}. \tag{12}$$

Hence

$$(ik)^2 \widehat{f}(k) = \sum_j \Delta_j e^{-ikt_j}, \tag{13}$$

up to boundary/window terms. Thus $|\widehat{f}(k)|^2$ has a k^{-4} envelope. Low rank changes the random sum and its prefactor, but not the generic envelope. The 1D validation must therefore use training transfer:

$$R_r(k, t) = \frac{|\widehat{f}_{r,t}(k)|}{|\widehat{f}^\star(k)|}. \tag{14}$$

This is the reason the main experiments focus on recovery slopes rather than raw random-network exponents in 1D.

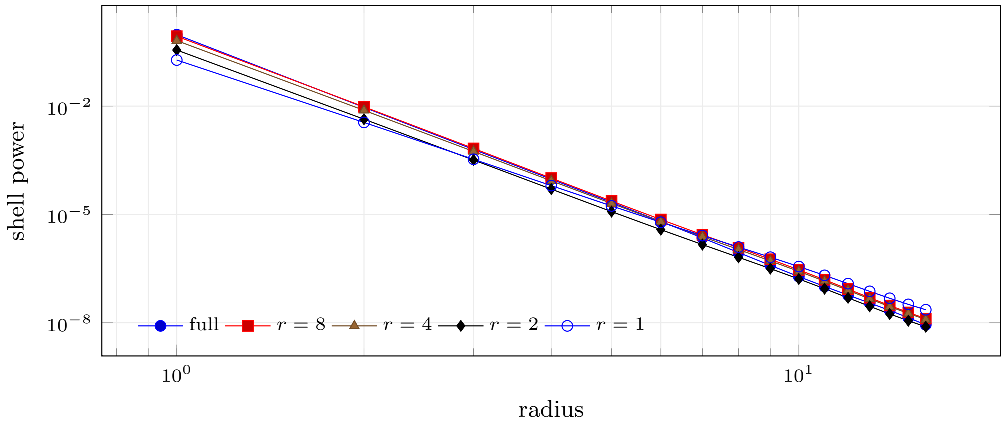

Figure 1: Shell spectra behind the slope plot. Low ranks preserve a visibly heavier relative high-frequency tail over the measured shell range.

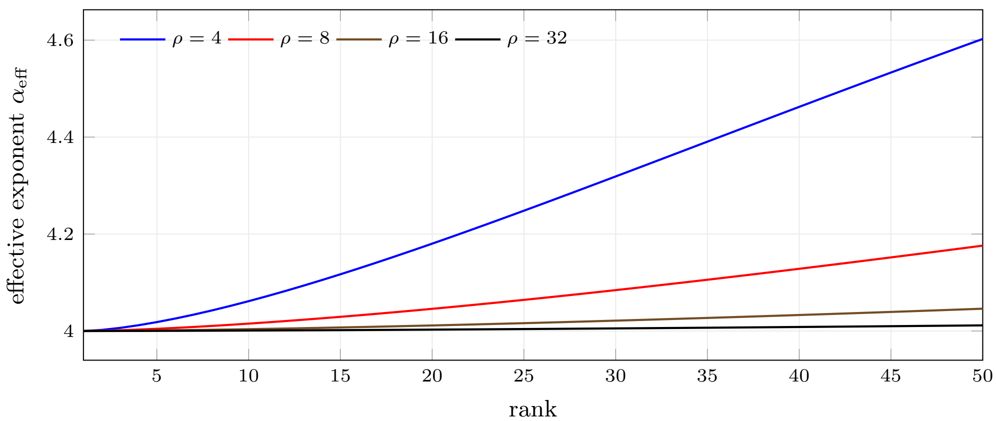

Figure 2: Resolution-dependent phase transition. High-codimension terms need larger moments to affect higher Fourier shells, so the rank at which the exponent detaches depends on the observed radius.

6 Detailed optimal-rank proxy

The main bound is

$$\mathcal{E}_r(t) \le Ar^{-2s} + Bd_{\text{eff}}(r)/n + \sum_k |\widehat{f}^\star(k)|^2 e^{-2t\lambda_r(k)}. \tag{15}$$

The minimizer satisfies a discrete balance condition: increasing rank by one is beneficial only if the approximation decrease is larger than the optimization/generalization increase,

$$A(r^{-2s} - (r+1)^{-2s}) \gtrsim \Delta_r \left[\sum_k |\widehat{f}^\star(k)|^2 e^{-2t\lambda_r(k)}\right] + \frac{B}{n}\Delta_r d_{\text{eff}}(r). \tag{16}$$

In our experiments, ranks 1 and 2 are dominated by approximation. Ranks much larger than 4 have a worse high-frequency transfer at the same training horizon. Rank 4 is the first point at which the approximation floor is low and the recovery slope remains flat.

7 Additional Fourier recovery plots

8 Sobolev/Fourier loss proxy

A Fourier-weighted Sobolev loss increases the value of high-mode recovery. In our proxy, increasing the Sobolev exponent shifts the risk curves in favor of ranks with flatter recovery slopes, but the approximation floor prevents

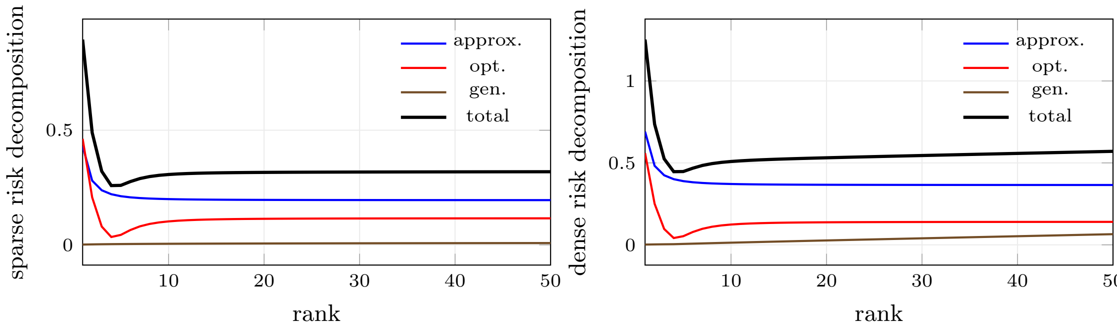

Figure 3: Approximation/optimization decomposition. The total proxy selects the same sweet spot as the experimental rank sweep.

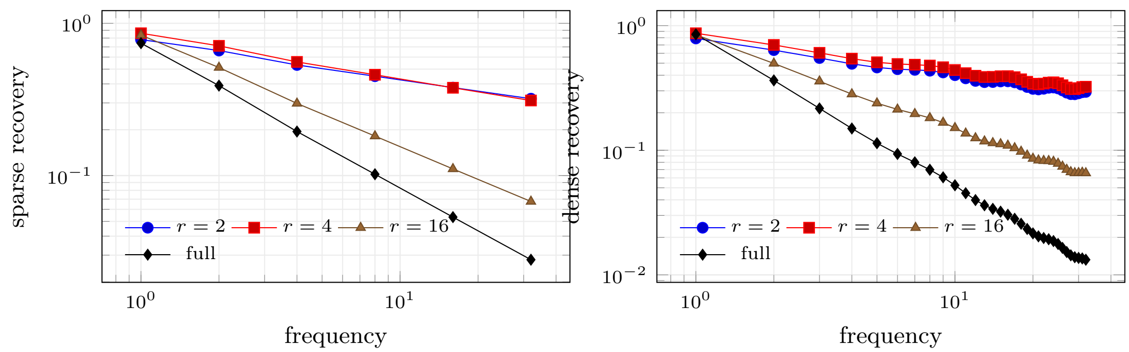

Figure 4: Mode-wise recovery for more ranks. Rank 2 is flatter but underfits; rank 16 starts to recover the full-rank spectral bias; rank 4 balances both.

the optimum from collapsing to rank 1.

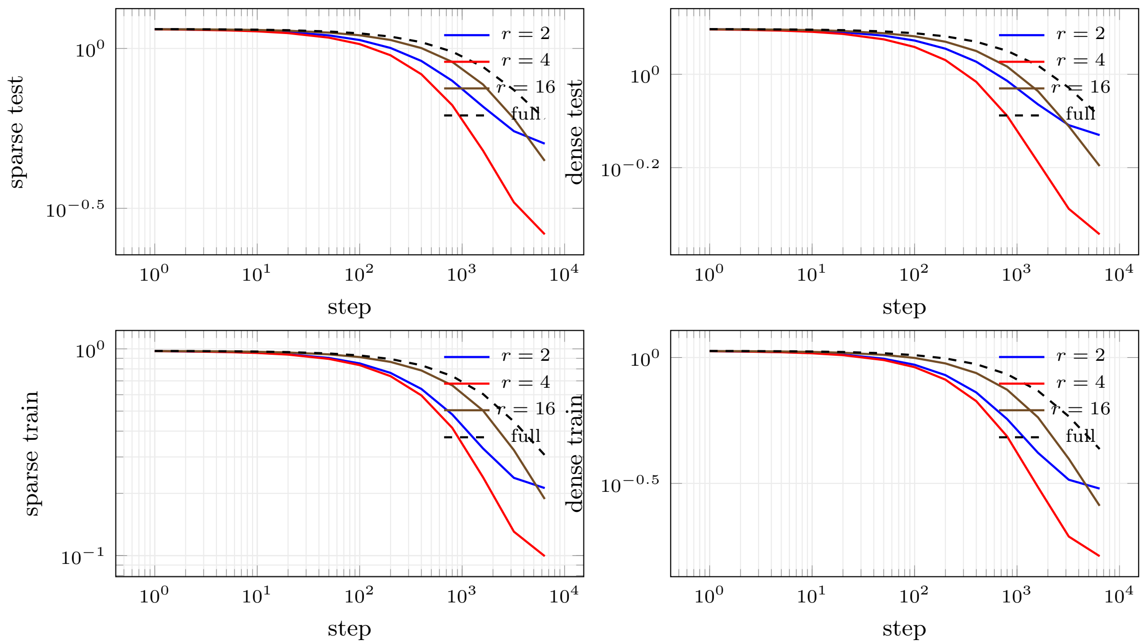

Figure 5: Training and test curves. The optimal rank is not merely a final-step artifact; it dominates over a broad finite-horizon window.

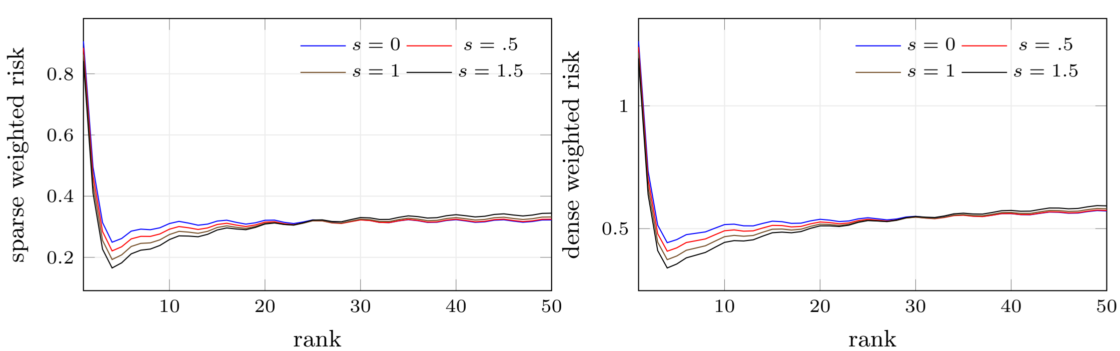

Figure 6: Sobolev/Fourier objective proxy. Fourier-weighted training strengthens the case for intermediate ranks because the high-mode transfer matters more.

9 Archived width-1024 multidimensional Fourier sweeps

Before focusing the final story on the 1D Fourier transfer law, we also ran finite-width sweeps on Fourier mixtures in dimensions $d \in \{1, 2, 5, 10, 20\}$ with sparse and dense spectra. These plots are included here to preserve the full experimental trail. They show the same qualitative message but with substantially more variance: rank is a useful spectral knob, yet the optimal rank depends strongly on dimension, sampling density, and mixture sparsity. This motivated the cleaner width-2^{15} one-dimensional Fourier/kernel experiment in the main paper.

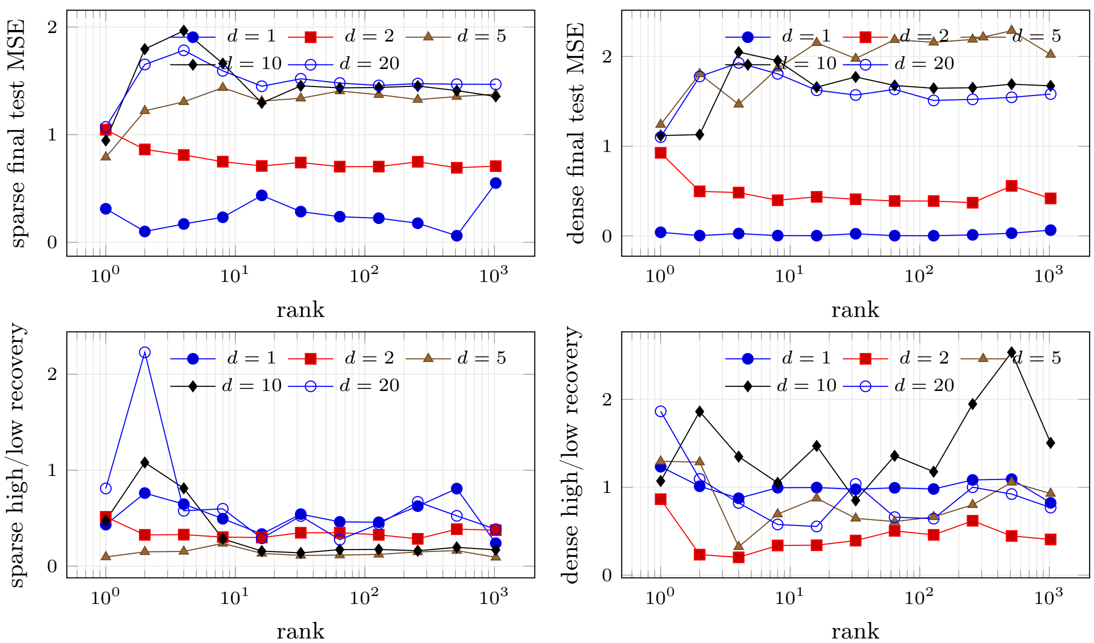

Figure 7: Earlier finite-width Fourier sweeps across dimensions. The signal is noisier than the width-2^{15} 1D kernel experiment, but it supports the same view: rank changes both finite-budget MSE and high-frequency recovery, and the best rank is task dependent.

Mixture	dim	best MSE rank	best MSE	best high/low rank
Sparse	1	512	0.060	512
Sparse	2	512	0.692	1
Sparse	5	1	0.790	8
Sparse	10	1	0.946	2
Sparse	20	1	1.071	2
Dense	1	128	0.0019	1
Dense	2	256	0.370	256
Dense	5	1	1.241	1
Dense	10	1	1.118	512
Dense	20	1	1.101	1

Table 1: Best ranks from the earlier finite-width multidimensional runs. These are not used as the main validation because dimension and sample sparsity introduce much larger variance, but they are useful context.

10 Implementation notes

The released script writes all CSV files used by the paper. It includes: CPWL geometry summaries; theory phase-transition curves; width-2^{15} rank sweeps; full endpoint lines; mode-wise recovery; train/test learning curves; and Sobolev-weighted proxy risks. The full endpoint is not a materialized dense 32768×32768 matrix. It is the dense/full transfer endpoint of the Fourier/kernel surrogate. This is the computationally meaningful object at this width and avoids a multi-gigabyte matrix that would be irrelevant to the Fourier transfer calculation.